

Functional Requirements for TechReserve (Borrower Side)

1. Authentication and Access Control

Module Capability	Detailed Specification
Institutional Login	• The system shall integrate with FEU Tech's Google Workspace via OAuth 2.0 to allow students, faculty, and staff to log in using their @feutech.edu.ph email addresses. • For manual logins, the system shall enforce a strict password policy (minimum 8 characters, alphanumeric, special characters) and account lockout after five failed attempts.
Session Management	• The system shall automatically terminate the user session after 30 minutes of inactivity. • All login, logout, and session expiration events shall be logged with timestamp and user reference.

2. Borrower Profile, Role, and Account Module

Module Capability	Detailed Specification
User Profile Details	• The system shall maintain the following user profile attributes: Last Name varchar 50 First Name varchar 50 Email Address varchar 100 Contact Number varchar 20 Last Login Timestamp datetime
Account Controls	• User access shall be restricted to self-owned data only. The system shall ensure that the users can: • View only their own profile • Submit reservation requests • View, modify, or cancel their own reservations • Upload required documentation for their reservations • View reservation statuses and history

Permissions	• The system shall allow users to: • Create new reservation requests • Select equipment and quantity • Specify event details and activity type • Upload required supporting documents • The system shall prevent users from: • Viewing reservations of other users • Viewing equipment allocation decisions of FO • Modifying equipment records • Approving or rejecting reservations • Accessing analytics configuration or raw analytics data • Viewing internal audit logs • Accessing system configuration or user management modules
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3. Equipment & Inventory Management Module

Module Capability	Detailed Specification
Equipment Listing	• The system shall display all available equipment including: • Equipment Name • Category • Available Quantity • Operational Status (Active/Inactive) • The system shall allow filtering by: • Category • Availability
Real-Time Availability Check	• The system shall validate equipment validity based on: • Requested date and time • Existing approved reservations • Current equipment state • The system shall prevent selection of quantities exceeding available inventory.

4. Reservation Submission Module

Module Capability	Detailed Specification

Unified Booking Interface	• The system shall allow requesters to select venues and required equipment in a single digital form.
Create Reservation	• The system shall allow users to submit a reservation request containing: • Reservation ID (Auto-generated format: TR-YYYY-NNN) • Organization Name • Requested Equipment • Quantity • Event Date and Time • Purpose/Activity Type • Submission Timestamp • Required Document Upload • New reservations shall initially be marked as: • Pending Review
Reservation Validation	• Before submission, the system shall: • Validate required fields • Check equipment availability • Prevent duplicate overlapping requests by the same organization • Display a confirmation summary prior to final submission

5. Reservation Tracking Module

Module Capability	Detailed Specification
Reservation Status Monitoring	• The system shall allow users to track reservation lifecycle states: • Pending Review • Approved • Rejected • Prepared • Deployed • Completed • Cancelled • Each status change shall include: • Timestamp • Status description • Rejection reason (if applicable)

Reservation Timeline View	• The system shall provide a visual status timeline showing: • Submission → Review → Approval → Preparation → Deployment → Return Confirmation
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6. Reservation Modification & Cancellation Module

Module Capability	Detailed Specification
Pre-Approval Modification	• The system shall allow users to: • Edit reservation details before approval • Cancel reservations before approval • Modification shall reset status to Pending Review
Post-Approval Restrictions	• After approval: • Reservation details shall be locked • Modification requests must be submitted to FO Admin • Cancellation requires administrative confirmation • Upon approved cancellation, equipment inventory shall automatically update.

7. Equipment Release & Return Visibility Module

Module Capability	Detailed Specification
Release Notification	• The system shall notify users when equipment status changes to: • Prepared • Ready for Release • Deployed

Return Confirmation	• After return confirmation by FO: • Status shall update to Completed • Users shall receive notification • Any recorded damage shall be visible to the users
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8. Home and Notifications Management

Module Capability	Detailed Specification
Home Landing Page	• The system shall display advisory notices such as: • “High demand period approaching” • “Limited equipment availability expected next week”
Automated Alerts	• The system shall notify users for: • Successful reservation submission • Reservation approval • Reservation rejection (with reason) • Upcoming reservation reminder (24 hours prior) • Equipment ready for release • Return confirmation • Notification delivery methods: • In-app notification • Email
Readiness Risk Indicator	• For approved reservations, the system may display: • Readiness Risk Level (Low / Medium / High) • Advisory message: “Preparation workload is high for selected dates.” • Risk indicators shall not prevent reservation submission.

9. Reservation History & Error Handling Module

Module Capability	Detailed Specification
Reporting	• The system shall allow users to: • View Historical reservations • View past approval/rejection decisions

Error Handling	• The system shall display clear, user-friendly error messages for: • Invalid form input • Equipment unavailability • Session expiration • Network errors • Users shall have a “Retry” option for failed submissions when applicable.
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Non-Functional Requirements for TechReserve (Borrower Side)

Category	Detailed Specification
Performance	• Reservation submission shall complete within 2 seconds. • Equipment availability validation shall complete within 2 seconds. • The system shall support simultaneous reservation requests from multiple users.
Usability	• The system shall feature a responsive web design that adapts seamlessly to desktop monitors • Forms shall provide inline validation and contextual tooltips. • Critical actions (submission, cancellation) shall require confirmation dialogs.
Reliability and Availability	• The system shall maintain an uptime of 99.5% during critical university operating hours (6:00 AM to 10:00 PM).
Security	• All data transmissions must be encrypted using TLS/SSL (HTTPS) protocols. • The system shall comply with the Philippine Data Privacy Act of 2012 (RA 10173), ensuring that user data (student IDs, contact information) is securely encrypted at rest. • Database backups shall be executed automatically every 24 hours, utilizing a 30-day rolling retention policy stored on secure cloud infrastructure.

	• Users shall only access their own reservation records.
Scalability	• The database architecture shall be designed to efficiently index and store up to 5 years of detailed transactional reservation logs, as this volume of historical data is critical for continuously training and refining the Time Series forecasting model.